space flight program remained secure. The official line was that the momentum behind its continuation and expansion was unstoppable. Many observers felt, though, that NASA's human space flight program was lacking in direction and focus. The sense of drift ended in January 2004, when President Bush announced his "vision for space exploration." There was to be a return to the Moon to establish a permanent base, and a future journey onward to Mars.

New direction

This fresh clarity of goal was welcomed by NASA, but required a substantial change of direction. It had long been obvious that the aging Shuttle fleet would need to be phased out, but the assumption had always been that a Shuttle replacement would be produced. This time it would be a fully reusable launch system, flying under its own power into space and back. The pursuit of this concept, which had originally inspired the Shuttle program had been renewed in 1994 with a directive from President Bill Clinton. It brought a new crop of experimental X-planes with the ambition of creating a "space liner" that would ferry people back and forth between bases on Earth and the ISS – and transform the economics of journeys into space. From 2004, however, NASA switched to development of a Crew Exploration Vehicle to go to the Moon and beyond – what became the Orion spacecraft. Replacing the Shuttle was no longer a priority. The "space liner" was left to private companies to develop instead.

NASA'S HYPER-X
The X-43A scramjet was created by Micro Craft for NASA to explore the possibility of hypersonic flight at speeds up to Mach 10 using an air-breathing engine. One of the possible uses for such an aircraft would be as a reusable space launch vehicle.

Human versus robotic

Many scientists remained openly sceptical of the value of human, as opposed to robotic, ventures into space. The achievements of robotic space probes and satellites in the first 50 years of the space era were indeed impressive. They had increased scientific knowledge of the Universe to a degree that human missions could not begin to match. Since the 1960s, solar observatories, astronomical observatories, and geophysical observatories have orbited the Earth, providing a vast quantity of new information on phenomena such as black holes and solar flares, with no need of scientist-astronauts to crew them. Nor does exploration of the solar system require humans to travel through space. The achievements of robot planetary explorers have been extraordinary. Thanks to NASA's Magellan spacecraft of 1989, for example, we have an accurate map of the surface of Venus, permanently hidden by cloud but revealed by radar. Since the Viking mission of 1976, we know what the surface of Mars looks like close up and have analyzed the planet's soil and atmosphere. Voyagers 1 and 2 gave new insight into Saturn and remote Uranus, and Galileo in 1996 produced many surprising revelations about Jupiter and its moons. This process of automated satellite exploration continued into the 21st century with projects from the series of Mars Rovers to the Juno mission and the New Horizons mission to Pluto.

The practical impact of automated satellites on everyday human life has been all-pervasive. They have, among other things, revolutionized communications, meteorology, cartography, and navigation. Thanks to satellites an individual can, at little cost, know his or her location on the Earth's surface at all times with great precision, and contact another person instantaneously almost anywhere on the globe.

The impetus to human space flight comes, however, from a different impulse to either scientific research or practicality. Of course, international competition has always been an important driving force. China's first human space flight under the Senzhou program in 2003 and the Chinese launch of the Tiangong-1 space station in 2011 gave a definite renewed competitive stimulus to the American space effort. But more important still was the undying human sense of adventure. This was evident in the continued push toward developing space tourism, with entrepreneurs such as Virgin's Richard Branson and Amazon's Jeff Bezos backing projects for commercial flights into space for paying passengers. The idea of walking on the surface of Mars or of establishing a permanent base on the Moon appealed to the human imagination in a manner that was probably irresistible. Despite achieving extensive exploration of Mars with robotic missions, NASA remained committed to landing humans on the planet by around 2035 – ultimately just to prove that it could be done. Space travel had been an inspirational dream since the days of Tsiolkovsky and Goddard, and so it remains in the 21st century.

PLANETARY EXPLORATION
Impressions of the future: below, a remote-controlled NASA airplane to fly survey missions on Mars; right, an orbiter powered by a tether – a long wire that generates an electric current as it passes through a planet's magnetic field.